

Beatriz De Moraes Caldas Lopes Manso

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Bioinformatics scientist with a PhD in Bioinformatics and hands-on experience building scalable computational pipelines, deep learning models, and variant-interpretation tools. Skilled in translating complex genomic and transcriptomic data into robust, reproducible software using Python, R, Snakemake, and modern machine learning frameworks. Background spans academic research, open-source software development, and full-time teaching leadership, with a focus on applying computational methods to real-world biomedical and clinical problems.

Technical Skills

- Programming & Tools: Python, R, C++, Bash, Git, Snakemake, Linux, Cloud Computing, HPC
- Machine Learning: Deep Learning (CNNs, ANNs), Scikit-learn, Keras, TensorFlow
- Genomics & Bioinformatics: RNA-seq, Microarray, Variant Annotation, GWAS, limma, DESeq2
- Other: Web Development, Project Management, Technical Writing

Research Interests

- Genomic data integration & transcriptomic analysis
- Deep learning for gene expression prediction (ANNs, CNNs)
- Rare diseases and cancer genomics
- Variant annotation & interpretation pipelines

Education

PhD in Bioinformatics

University of West London, UK | Sep 2022 – Dec 2025

Focus: Genetic basis of Chiari Malformation Type I (CM-I) and its potential correlation with brain cancers using gene expression data and deep learning techniques.

Methods: Gene expression analysis, machine learning CNN/ANN development, GWAS, pathway enrichment, multi-omics integration

MSc in Bioinformatics – Distinction

University of West London, UK | Feb 2021 – Aug 2022

BSc in Biomedical Sciences

CESPU, Porto, Portugal | Sep 2016 – Aug 2020

Erasmus Program

Universitat de Valencia, Spain | Jan 2019 – Jul 2019

Teaching Experience

Course Leader, MSc Bioinformatics & Module Leader, BSc Modules

University of West London | Sep 2025 – Present

- Lead curriculum design, delivery, and quality assurance for the MSc Bioinformatics programme
- Serve as module leader for two BSc modules, overseeing content design, delivery, and assessment
- Manage teaching staff and coordinate timetabling, liaising with programme leadership on course development
- Continue mentoring and supervising MSc and BSc students on computational biology coursework and projects

Teaching Assistant – MSc Bioinformatics

University of West London | Sep 2022 – Aug 2025

- Delivered lectures and practical sessions for modules including Bioinformatics and Genome Analysis and Functional Genomics
- Designed and delivered practical lab materials and exercises, guiding students through hands-on computational biology work

- Marked assessments and provided feedback to MSc and BSc students, fostering strong engagement and understanding

Professional Experience

Bioinformatics Software Developer – Internship

OakVar (Remote) | Jan 2023 – Aug 2023

- Built Snakemake wrappers for OakVar's variant annotation platform, enabling streamlined, reproducible genomic workflows for clinical variant interpretation
- Collaborated with bioinformatics developers to improve reproducibility, scalability, and modularity of workflows
- Contributed to open-source development and documentation

Web Content Manager

Noveris, LDA, Porto | Oct 2018 – Nov 2021

- Led optimization of company website through performance analytics, content strategy, and branding
- Managed internal and external projects, coordinating cross-functional teams
- Analyzed site analytics to drive continuous UX and performance improvements
- Oversaw full project lifecycle: content creation, brand development, and marketing delivery
- Partnered with IT to implement cybersecurity practices and strengthen project lifecycle management

PhD Research Overview

PhD Thesis Project

- Developed interpretable deep learning models (ANNs, CNNs) for classification and prediction of gene expression in Chiari Malformation I using gene expression datasets
- Conducted comparative genomic analysis of CM-I and brain cancer patients using public datasets
- Integrated machine learning pipelines with bioinformatics tools for robust and interpretable results
- Applied systems biology and statistical approaches to explore GWAS/WES data and pathway correlations

Academic Presentations

- Internal PhD Seminars, University of West London: Presented findings on CM-I gene expression and CNN-based predictions
- Manuscript under review in peer-reviewed journal

Additional Information

- Eligible to work in the UK
- Fluent in English and Portuguese